Software Test Plan (STP) — Real-Time Collaborative Virtual Whiteboard

Project: Real-Time Collaborative Virtual Whiteboard

Team: Team 11 — Rachit S Mane, P Chinni Krishna Reddy, Pranav V Menon, Prithvi W

Version: 1.0

Date: 11-10-2026

Status: Final for submission — team sign-off pending (see §15). Execution 12-10-2026 (uncommitted working tree on base commit 82af885): 32 test cases Pass, 2 Blocked (TC-021: stable Firefox not installed at that time; TC-109: no deployment), none Fail; details in each test case and in `docs/IMPLEMENTATION_STATUS.md`. Before this execution no test had been run. TC-021 was executed manually on 17-10-2026 (installed Firefox 158 and Chrome by hand; Chrome and Edge automated) and set to **Pass** with limitations accepted by the team after a two-member review (TC-021 record), so 33 test cases are Pass and 1 Blocked (TC-109). Final regression 02-11-2026 on `main` after PR #33 (§14, `docs/reports/RELEASE_READINESS_2026-11-02.md`): no failure; statuses unchanged.

Revision history

Version	Date	Author	Change summary
0.1	10-10-2026	Pranav, Prithvi (draft)	First complete draft for team review
0.2	10-10-2026	Pranav, Prithvi (draft)	Status definitions and RTM mapping; version-history tests aligned with SRS §4.3; TC-018 ownership transfer; new TC-022 (board size limit) and TC-023 (member roles); reproducible performance procedures; dependency field in every test case
0.3	10-10-2026	Pranav, Prithvi (draft)	TC-022 uses the BSON size measure; TC-005 notes unit-level verification of a rejected clear; RTM columns as in the SRS template
0.4	10-10-2026	Pranav, Prithvi (draft)	TC-022 rewritten with measured sizes M1–M4 and a deterministic generator; new TC-024 (version write failure, interrupted restore recovery); stale-permission checks added to TC-018 and TC-103; TC-107 message boundary in bytes; implementation status of all test assets stated
1.0	11-10-2026	Pranav, Prithvi	Final for submission: TC-024 (restore recovery, idempotent retry) and new TC-025 (clear recovery); targets referenced as specified in SRS §5; §11 sign-off stated as a precondition; TC-024 and TC-025 cover every failure point of SAD §3.6; TC-104 checks the log fields of SAD §4.4; TC-108 tests an unregistered e-mail; TC-104 and §5.1 list `subjectId`

1. Introduction

Purpose: This document defines the test plan for the Real-Time Collaborative Virtual Whiteboard v1.0. It describes the objectives, scope, strategy, environment, schedule, responsibilities, test cases and traceability used to verify the requirements of the SRS.

Scope: Testing covers canvas editing, real-time synchronisation and recovery, version history, accounts and boards, share links and roles, the performance and capacity targets, browser compatibility and usability, and the security requirements of SRS §5.1. Items excluded from testing are listed in §4.

References: Team 11 SRS v1.0; Team 11 SAD v1.0; master RTM (`docs/traceability/rtm.md`); Decision Log (`docs/DECISIONS.md`); course brief "Mini-Project Deliverables Part-1"; supplied STP template.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), RTM (Requirements Traceability Matrix), TC (test case), TB-STD (standard test board, §6), SUT (system under test), E2E (end-to-end), p95 (95th percentile). Other terms as in SRS §1.4.

2. Test Items

Test item	Component (SAD §3.4)
Whiteboard client: canvas, toolbar, panels, dialogs	COMP-01
Client synchronisation and reconnection logic	COMP-02
REST API (accounts, boards, links, members, versions)	COMP-03
Authentication service	COMP-04
Board and access service, permission policy, share links	COMP-05
Real-time collaboration server (Socket.IO events, sequencing, presence)	COMP-06
Version history service	COMP-07
Persistence layer and database indexes	COMP-08
Logging and monitoring, health endpoint	COMP-09

3. Features to be Tested

Features mapped to SRS requirement IDs and test cases (full mapping in §13):

Feature	Requirement IDs	Test cases
Freehand drawing, colour and width	FR-001, FR-002, FR-020	TC-001
Shapes and text	FR-003, FR-004, FR-020	TC-002
Object editing and erasing	FR-005, FR-006	TC-003, TC-004
Owner-only clear with pre-clear version, interrupted-clear recovery; board size limit	FR-007, FR-030, FR-008	TC-005, TC-022, TC-025
Ordering and concurrent edits	FR-021, FR-022, FR-023	TC-006, TC-007
Joining, reconnection, rejection, persistence	FR-024, FR-025, FR-026, FR-027, NFR-004	TC-008, TC-009, TC-010, TC-011

Feature	Requirement IDs	Test cases
Version history: automatic versions and no-change rule, list, preview, restore, retention	FR-030 to FR-035, NFR-005	TC-005, TC-012, TC-013, TC-014, TC-024, TC-102
Accounts and boards	FR-040 to FR-044	TC-015, TC-016, TC-102
Share links, membership, roles, ownership transfer, presence	FR-045 to FR-052	TC-017, TC-018, TC-023, TC-101, TC-102, TC-103
Performance and capacity	NFR-001, NFR-002, NFR-003	TC-019, TC-020
Compatibility, 1280 px viewport and usability	NFR-006, NFR-007	TC-021
Security requirements	SEC-REQ-001 to SEC-REQ-008	TC-101 to TC-109 (see §5.1)

4. Features Not to be Tested

Item	Reason
Candidate enhancements (undo/redo across users, live cursors, export, chat, advanced version comparison, mobile optimisation)	Not in version 1.0 scope (SRS §2.2).
Behaviour on unsupported browsers and mobile devices	Outside NFR-006.
Horizontal scaling across several server instances	Out of scope (SRS §2.5).
Internal correctness of third-party libraries (React, Express.js, Socket.IO, MongoDB, bcrypt, JWT library)	Assumed tested by their maintainers; only our use of them is tested.
Performance over the public Internet or with artificial network delay	Targets are defined for the local test environment (SRS §5, D7b).
Long-term availability (uptime percentage)	No availability requirement in version 1.0.
Formal penetration test by an external party and formal WCAG audit	Beyond course resources; replaced by the security test cases in §5.1 and the usability checklist in TC-021.

Note (02-11-2026). Two functions were implemented although they are outside the version 1.0 requirement baseline: PNG export (issue #16; listed above as a candidate enhancement) and board rename (issue #13). They have no requirement ID, no test case in this plan and no RTM row; their scope decision is pending (SRS §2.7). They are nevertheless covered by automated tests outside this plan's test-case set: `client/tests/exportImage.test.js`, `server/tests/integration/boards.test.js` and `e2e/boards.spec.cjs` (and the release smoke test `e2e/smoke.spec.cjs`). If they are brought into scope, new test cases and RTM rows are added in the same change.

5. Test Approach / Strategy

Levels:

- **Unit tests:** permission policy (every cell of the FR-051 matrix), payload schemas, sequencing and last-write-wins logic, version triggers and retention, token issuing and verification, client reconciliation (pending, confirm, rollback).
- **Integration tests:** REST endpoints against a test database (API-02, API-03); Socket.IO event flows with several scripted clients (API-01); interaction of COMP-07 with COMP-06 during restore.

- **System (end-to-end) tests:** browser-driven scenarios with two or more real browser sessions for the functional test cases (TC-001 to TC-018, TC-022, TC-023); TC-024 and TC-025 are integration tests with fault injection.
- **Acceptance tests:** the suites listed in SRS §6, run on the release candidate and reviewed by all four members.

Types:

- Functional testing (positive and negative) of every functional requirement.
- Concurrency testing with scripted clients that submit operations at the same time (TC-006, TC-007).
- Recovery testing by disconnecting clients, restarting the server and injecting faults during version writes and restores (TC-009, TC-011, TC-024, TC-025).
- Performance and capacity testing with a latency harness of 10 scripted clients (TC-019) and browser timing (TC-020).
- Security testing: authorisation matrix, crafted messages, link lifecycle, token handling, input validation, rate limiting, log inspection (§5.1).
- Compatibility, viewport and usability checklist testing (TC-021).
- Boundary testing of limits (TC-001, TC-002, TC-015, TC-016, TC-022, TC-105, TC-107).
- Regression testing: the automated unit and integration suites run in CI on every Pull Request (configured: `app-ci` — lint, unit, integration, build and Chromium E2E — and `docs-check` run on every Pull Request and push to `main` since 01-11-2026, DECISIONS.md D16) and locally before every push.

Test design techniques: equivalence partitioning and boundary values for limits (stroke width 1/20/0/21, text 500/501 characters, password 7/8/72/73 characters, 65,536-byte message size, 15,728,640-byte board content size); decision tables for the permission matrix; state transitions for share links (active → expired / revoked) and connections (connected → disconnected → resynchronised).

Entry criteria (status at the execution of 12-10-2026: build, health check, test data and test-case review met; the team sign-off of SRS, SAD and STP v1.0 was **not** met — the execution was run before sign-off and must be confirmed or repeated after it, §11): SRS, SAD and STP v1.0 signed off by the team; build deployed in the test environment (§6) and the health check returns `ok`; test data (TB-STD, accounts) loaded; test cases reviewed by a second team member.

Exit criteria: 100% of planned test cases executed; all High-priority test cases passed; no open defect of severity Critical or High; every requirement in the RTM has status P or an approved, documented exception; test summary report reviewed by all four members.

Defect severity: Critical — data loss, security breach or crash; High — a requirement fails with no workaround; Medium — a requirement fails with a workaround; Low — cosmetic.

5.1 Security Validation

Every security requirement of SRS §5.1.2 is validated by at least one test case. Security test cases TC-101 to TC-109 are specified in Appendix A (drafted by the security lead, reviewed by the STP lead). Historical (v1.0): all were planned and none had been executed. Status after the execution of 12-10-2026 (uncommitted working tree on base commit 82af885; see `docs/IMPLEMENTATION_STATUS.md`): TC-101 to TC-108 Pass; TC-109 Blocked (no deployment).

Security requirement	Validation activity	Method	Test case(s)
SEC-REQ-001 Password policy and hashing	Boundary tests on password length; inspection of stored user records and logs	Automated (API) + database inspection	TC-105
SEC-REQ-002 Tokens, expiry, logout	Requests, socket connections and operations with missing, tampered, expired, logged-out and wrong-board tokens (session and guest)	Automated (API, Socket.IO)	TC-106
SEC-REQ-003 Server-side authorisation	Full permission matrix over REST (including ownership transfer and restore); crafted Socket.IO messages from Viewers and guests, including forged `role`/`boardId` fields; access to other boards' IDs (IDOR, 404 rule)	Automated (API, scripted Socket.IO clients)	TC-101, TC-102
SEC-REQ-004 Share-link tokens	Token length/alphabet check over 1,000 generated tokens; database holds only hashes; expired and revoked links rejected	Automated + database inspection	TC-103, TC-104
SEC-REQ-005 Input validation and XSS	Malformed, out-of-range and oversized (65,537-byte) messages, as distinct from the board content limit of FR-008 tested in TC-022; text containing `<script>` and HTML markup rendered in the browser	Automated (Socket.IO, API) + browser check	TC-107
SEC-REQ-006 Rate limiting	Six failed logins in 15 minutes for a registered and for an unregistered e-mail address (identical status, error code and message; only `requestId` differs); 100 operations in one second from one connection	Automated	TC-108
SEC-REQ-007 Transport security	Configuration review and TLS connection check of the deployed instance; HTTP → HTTPS redirect	Manual inspection	TC-109
SEC-REQ-008 Security logging without secrets	Run the security suite, then check that each required event type has an entry with event, timestamp, request ID, outcome and reason, actor or `null`, board or `null` and affected account (`subjectId`) or `null` as defined in SAD §4.4 (including a failed login with an unknown and with a registered e-mail); search the log for passwords, tokens, `Authorization` headers, entered e-mail addresses and board text	Automated log scan	TC-104

Additional checks during security validation:

- **Dependency audit:** `npm audit` on client and server before release; no known Critical vulnerabilities may remain unresolved without a documented reason.
- **Secrets check:** the repository is searched for credentials, tokens and connection strings before every release; test accounts use generated passwords that are never committed.
- **Local exemption:** the local test environment uses plain HTTP and WS; SEC-REQ-007 can be verified only on a deployed instance (TC-109). **No deployment exists at present**, so TC-109 is "Blocked" (reason: no deployment, §13) and SEC-REQ-007 stays at RTM status N; it is never reported as passed without a deployed test.
- **Status honesty:** a security requirement is verified only when all its linked tests have status Pass (§13). Documented but unexecuted tests verify nothing.

6. Test Environment

Hardware: one team laptop as the server host (application server and MongoDB); up to two additional laptops on the same local network for multi-browser tests. Hardware specifications are recorded in the test summary report with every performance result.

Software: Node.js LTS, MongoDB Community (local), current stable Google Chrome, Mozilla Firefox and Microsoft Edge (versions recorded at test time), build of the system under test identified by Git commit hash.

Browser builds used for automated runs (added 02-11-2026). `npm run test:e2e` uses Playwright's bundled Chromium (also in CI). `npm run test:e2e:browsers` uses the installed **stable Google Chrome** and **stable Microsoft Edge** (Playwright channels) and Playwright's own **Firefox** and **WebKit** builds. Playwright's Firefox is a patched build, **not** the stable Mozilla Firefox release that NFR-006 names; Playwright's WebKit is the Safari engine, **not** Apple Safari. Evidence for the installed stable Mozilla Firefox comes from manual runs (TC-021, Firefox 158) and from Playwright's experimental WebDriver BiDi driver (Firefox 157 on 16-10-2026, Firefox 159 on 02-11-2026), which cannot automate every step (no `dblclick`, no download events, no network outage). Apple Safari is not a target browser of NFR-006 and has not been tested.

Tools (DECISIONS.md D15): Jest (unit tests), Supertest (REST integration tests), socket.io-client scripted clients (Socket.IO integration, concurrency and latency harness), Playwright (browser end-to-end tests and load timing), browser developer tools (network throttling/offline simulation), MongoDB shell or Compass (database inspection), GitHub Issues (defect tracking), GitHub Actions (CI: `app-ci`, `docs-check`).

Implementation status of test assets: the scripts, harnesses, seed data, fixtures, accounts and hooks named in this plan exist and were used in the execution of 12-10-2026 (Appendix A, `docs/IMPLEMENTATION_STATUS.md`). Historical v1.0 note: when this plan was written none of them existed; they were planned deliverables of the implementation phase and blockers for executing the corresponding test cases.

Test hooks and configuration (test builds only, SAD §4.1.6): `board-ready` performance mark; `window.__wbTest.exportState()` and `submitRaw()`; `VERSION_INTERVAL_MS` (periodic interval for tests — the configured value is recorded in every result, and the production value is 5 minutes, D11a); `LINK_EXPIRY_OVERRIDE_SEC`; `FAULT_INJECT` for TC-024 and TC-025; a separate test JWT secret in `.env.test`. None of these is enabled in a deployed system.

Performance measurement procedure (TC-009, TC-013, TC-019, TC-020): the definitions of SRS §5 apply — start and end points, one monotonic clock per run (scripted clients in one Node.js process; Playwright timings from the page's own performance timeline), nearest-rank percentiles, warm-up excluded, rejected operations excluded from latency samples and reported separately, missing or duplicate sequence numbers detected per client, and final states compared with a fresh `board:join`. Raw samples and summaries are stored as CSV files under `test-results/` with the Git commit, date, hardware, operating system, Node.js, MongoDB and browser versions, and are compared with the targets of SRS §5.

Test data:

- **TB-STD** (DECISIONS.md D7a): 300 freehand strokes of at most 200 points each, 50 shapes and 20 text objects, generated by a seed script with a fixed random seed so every run uses identical content.
- Test accounts: one Owner, two Editors, one Viewer, plus guest sessions; passwords are generated by the seed script at run time and are not stored in the repository.
- Boundary data: stroke widths 0, 1, 20, 21; text of 500 and 501 characters; passwords of 7, 8, 72 and 73 characters; a 65,537-byte message (M1, just above the 65,536-byte limit); board-filling strokes produced by `gen-board-fill` with recorded M1–M4 sizes (TC-022); text containing `<script>alert(1)</script>`.

Environment conditions for performance tests (SRS §5, D7b): server and database on one laptop, clients on the same machine or LAN, no artificial delay, no other heavy applications running.

7. Test Schedule

The course brief's ETA (28-09-2026) has passed and the current deadline has not been confirmed, so the schedule is expressed relative to the team sign-off of the v1.0 documents (S). Calendar dates will be added once the deadline is confirmed.

Milestone	Planned timing
Test case design and review complete	S + 3 days
Test environment and TB-STD seed script ready	S + 5 days
Unit and integration test execution (continuous, in CI)	From first implementation PR
System and security test execution	After feature-complete build
Performance tests (TC-019, TC-020, timing parts of TC-009, TC-013)	Feature-complete build + 2 days
Acceptance review and test summary report	Before final submission

8. Test Deliverables

- Test Plan (this document)
- Test case specifications (Appendix A)
- Automated test scripts (unit, integration, end-to-end, latency harness) in the repository
- Test data seed script (TB-STD, accounts)
- Test execution logs and CI run records
- Defect reports (GitHub Issues)
- Updated RTM with status per requirement
- Test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead / STP lead	Pranav V Menon	Prepare plan, functional and non-functional test cases, coordinate execution and reporting
Security test lead	Prithvi W	Security test cases TC-101 to TC-109, security validation review, RTM maintenance
Test engineer / SRS lead	Rachit S Mane	Review test cases against acceptance criteria; execute version-history and canvas tests
Test engineer / SAD lead	P Chinni Krishna Reddy	Latency harness and Socket.IO integration tests; defect triage for COMP-02/COMP-06
Product owner (sign-off)	All four members jointly	Approve test results and readiness

Every member writes and runs tests for the features they implement (team rule 6) and reviews at least one other member's tests in Pull Requests.

10. Risks and Mitigation

Risk	Mitigation
NFR targets turn out infeasible when measured	Build the latency harness early; raise a change request for the affected NFR; report measured values honestly.
Timing measurements distorted by clock differences	Run all scripted clients in one process or on one machine.
Flaky concurrency tests	Fixed seeds, explicit waits on acknowledgements, repeated runs (10×) for timing tests.
CI not configured in time	Run the same commands locally before each push; record results in the PR description.
Shared laptop resources affect performance	Record hardware and background load; repeat runs; close other applications.
No deployed instance for TC-109	Report TC-109 as Blocked (reason: no deployment, §13), never as passed.

11. Assumptions & Dependencies

- The SRS v1.0 shall be signed off before test execution, and its requirement IDs shall remain frozen during testing. (At present the sign-off is pending, §15; the execution of 12-10-2026 therefore took place before sign-off, a recorded deviation.)
- The selected stack (D10) and tools (D15) are used; a change to either requires updating §6.
- Every team member has Node.js, MongoDB and the three browsers installed.
- The seed script and test accounts are available before system testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours.
- The build blocks more than 30% of the planned test cases.
- A Critical security defect is found (stop security sign-off until fixed).

Resume testing if:

- Blocking defects are fixed and verified.
- The environment is stable and the health check returns ok.
- Affected test cases are re-run from the beginning.

13. Test Case Management & Traceability

- Test case identifiers: TC-001 to TC-099 for functional and non-functional tests (owned by the STP lead), TC-101 to TC-199 for security tests (owned by the security lead). Identifiers are registered in `docs/traceability/id-registry.md` and never renumbered; a changed scope is recorded in the registry change log (e.g. TC-018 narrowed to ownership transfer in v0.2, with FR-049 moved to the new TC-023).
- Each test case lists the requirement IDs it verifies. The master RTM below maps every requirement to its test cases. `tools/check_trace.py` checks that every requirement has at least one test case, that every referenced test case exists, that the SRS, RTM and test cases agree on these links, and that RTM statuses agree with test-case statuses.
- Test case specifications are maintained in the repository (Appendix A) and changed only through reviewed Pull Requests. Execution results are recorded in the "Actual result" and "Status" fields; a result is recorded only from an actual run, together with the build (Git commit) and date.

Test-case status values

Status	Meaning
Not executed	The test has not been run on the current build (initial value).
Pass	The test was run and every pass criterion was met; "Actual result" records the evidence.
Fail	The test was run and at least one pass criterion was not met; a defect is raised.
Blocked	The test could not be run (e.g. missing environment, blocking defect, no deployment); the reason is recorded in "Actual result".

RTM status per requirement (derived; never set by hand)

RTM status	Rule
P	Every linked test case has status Pass.
A	At least one linked test case has status Fail (attempted, defect open).
N	Otherwise: at least one linked test case is Not executed or Blocked and none has failed. Blocked tests are named, with the reason, in the RTM "Comments" column.

A Blocked or Not executed test never counts as a pass. After the execution of 12-10-2026 the RTM status is P for every requirement except NFR-006 and NFR-007 (TC-021 Blocked) and SEC-REQ-007 (TC-109 Blocked), which remain N. After TC-021 was set to Pass on 17-10-2026 (with limitations, TC-021 record), NFR-006 and NFR-007 are P; only SEC-REQ-007 remains N.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
FR-001	Freehand stroke	SRS 4.1 / SD-01, API-01	COMP-01, COMP-02, COMP-06	TC-001	P	
FR-002	Colour and stroke width	SRS 4.1 / API-01	COMP-01	TC-001	P	
FR-003	Rectangle, ellipse, line	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-002	P	
FR-004	Text objects	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-002	P	
FR-005	Select, move, restyle, delete	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-003	P	
FR-006	Object eraser	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-004	P	
FR-007	Owner-only clear; pre-clear content kept; recovery	SRS 4.1 / SD-05, API-01	COMP-05, COMP-06, COMP-07, COMP-08	TC-005, TC-025	P	
FR-008	Board content size limit 15 MiB (M3)	SRS 4.1 / SD-01, API-01	COMP-01, COMP-06	TC-022	P	
FR-020	Broadcast accepted operations	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-001, TC-002	P	
FR-021	Server sequence numbers and ordering	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-006, TC-007	P	
FR-022	Independent edits preserved	SRS 4.2 / SD-01	COMP-06	TC-006	P	
FR-023	Same-object last write wins	SRS 4.2 / SD-01	COMP-02, COMP-06	TC-007	P	
FR-024	Full state on join	SRS 4.2 / SD-03, API-01	COMP-06, COMP-08	TC-008	P	
FR-025	Disconnect indicator and resync	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-009	P	
FR-026	Rejection feedback and rollback	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-010	P	
FR-027	Persist before broadcast	SRS 4.2 / SD-01	COMP-06, COMP-08	TC-011	P	
FR-030	Automatic versions, no-change rule	SRS 4.3 / API-03	COMP-07, COMP-08	TC-005, TC-012	P	
FR-031	Version list for all roles	SRS 4.3 / API-03	COMP-01, COMP-07	TC-012	P	
FR-032	Read-only version preview	SRS 4.3 / API-03	COMP-01, COMP-07	TC-012	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
FR-033	Owner-only restore as new version; recovery	SRS 4.3 / SD-02, API-03	COMP-05, COMP-07, COMP-08	TC-013, TC-024, TC-102	P	
FR-034	Restored state pushed to all	SRS 4.3 / SD-02, API-01	COMP-06, COMP-07	TC-013	P	
FR-035	Retain at least 20 versions	SRS 4.3 / API-03	COMP-07, COMP-08	TC-014	P	
FR-040	Register account	SRS 4.4 / API-02	COMP-03, COMP-04	TC-015	P	
FR-041	Log in and log out	SRS 4.4 / API-02	COMP-03, COMP-04	TC-015	P	
FR-042	Create board, creator is Owner	SRS 4.4 / API-02	COMP-03, COMP-05	TC-016	P	
FR-043	Dashboard of boards	SRS 4.4 / API-02	COMP-01, COMP-03, COMP-05	TC-016	P	
FR-044	Owner-only board deletion	SRS 4.4 / API-02	COMP-05, COMP-06	TC-016, TC-102	P	
FR-045	Create share link with role and expiry	SRS 4.4 / API-02	COMP-05	TC-017	P	
FR-046	Join via link (member or guest)	SRS 4.4 / SD-03, API-02	COMP-04, COMP-05	TC-017	P	
FR-047	Revoke share link	SRS 4.4 / SD-03, API-02	COMP-05, COMP-06	TC-103	P	
FR-048	Expired link rejected	SRS 4.4 / SD-03	COMP-05, COMP-06	TC-103	P	
FR-049	Change member role or remove	SRS 4.4 / API-02	COMP-05, COMP-06	TC-023	P	
FR-050	Transfer ownership to a member	SRS 4.4 / SD-04, API-02	COMP-03, COMP-05, COMP-06	TC-018	P	
FR-051	Permission matrix enforced	SRS 4.4 / SD-01, SD-02, SD-03, SD-04, SD-05	COMP-03, COMP-05, COMP-06	TC-101, TC-102	P	
FR-052	Participant presence list	SRS 4.4 / API-01	COMP-01, COMP-06	TC-017	P	
NFR-001	95% of updates visible within 500 ms	SRS 5 / SD-01	COMP-02, COMP-06, COMP-09	TC-019	P	
NFR-002	At least 10 concurrent participants	SRS 5	COMP-06	TC-019	P	
NFR-003	TB-STD loads within 3 s	SRS 5	COMP-01, COMP-03	TC-020	P	
NFR-004	Reconnect recovery within 5 s	SRS 5 / SD-01	COMP-02, COMP-06	TC-009	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
NFR-005	Restore visible to all within 3 s	SRS 5 / SD-02	COMP-06, COMP-07	TC-013	P	
NFR-006	Chrome, Firefox, Edge support	SRS 5	COMP-01	TC-021	P	TC-021 Pass with limitations (17-10-2026): manual in Firefox 158 and Chrome, automated in Chrome and Edge; Edge not checked by hand; the automated run predates the stroke-width tooltip fix
NFR-007	Role/status visible, labelled toolbar	SRS 5	COMP-01	TC-021	P	TC-021 Pass with limitations (17-10-2026): stroke-width tooltip defect found and fixed during TC-021, re-checked in Firefox only
SEC-REQ-001	Password policy and salted hashing	SRS 5.1.2 / API-02	COMP-04, COMP-08	TC-105	P	
SEC-REQ-002	Tokens required, expiry, logout	SRS 5.1.2 / SD-03, API-01, API-02	COMP-03, COMP-04, COMP-06	TC-106	P	
SEC-REQ-003	Server-side authorisation of every request/event	SRS 5.1.2 / SD-01, SD-02, SD-03, SD-04, SD-05	COMP-03, COMP-05, COMP-06	TC-101, TC-102	P	
SEC-REQ-004	Link token entropy, hashing, expiry	SRS 5.1.2 / SD-03, API-02	COMP-05, COMP-08	TC-103, TC-104	P	
SEC-REQ-005	Schema validation, size limit, no HTML	SRS 5.1.2 / API-01, API-02, API-03	COMP-01, COMP-03, COMP-06	TC-107	P	
SEC-REQ-006	Login and operation rate limits	SRS 5.1.2 / API-01, API-02	COMP-03, COMP-06	TC-108	P	
SEC-REQ-007	TLS 1.2+ in deployment	SRS 5.1.2	COMP-03, COMP-06	TC-109	N	TC-109 Blocked: no deployed HTTPS instance
SEC-REQ-008	Security event logging without secrets	SRS 5.1.2 / SD-04	COMP-03, COMP-04, COMP-05, COMP-06, COMP-07, COMP-09	TC-104	P	

Security objectives are traced through the security requirements that realise them (SRS 5.1.1):

Objective	Realised by	Test case(s)	Status (N/P/A)
SEC-OBJ-001	SEC-REQ-002, SEC-REQ-003, SEC-REQ-004, SEC-REQ-007	TC-101, TC-102, TC-103, TC-104, TC-106, TC-109	N
SEC-OBJ-002	SEC-REQ-003, SEC-REQ-005, FR-033	TC-013, TC-101, TC-102, TC-107	P
SEC-OBJ-003	SEC-REQ-006, SEC-REQ-008	TC-104, TC-108	P

14. Test Metrics & Reporting

Metrics collected:

- Percentage of planned test cases executed
- Percentage passed / failed
- Requirement coverage (requirements with RTM status P ÷ all requirements; see §13 for the derivation)
- Defects by severity, defect density (defects per component), defect ageing
- Performance results: p95 latency (NFR-001), maximum participants without errors (NFR-002), load, reconnect and restore times (NFR-003 to NFR-005) with hardware recorded

Reports:

- Execution status in each Pull Request (CI result and local test command output)
- Weekly execution summary during system testing
- Final Test Summary Report: results per test case, RTM status, open defects, measured performance values, and deviations from this plan

Every performance result is reported with its measured value, not only as pass or fail. Metrics of the 12-10-2026 execution: 34 test cases; 32 Pass (94%), 0 Fail, 2 Blocked (TC-109 not run at all; TC-021 run in Chrome and Edge only); requirement coverage 47 of 50 requirements with RTM status P; measured values in the TC-009, TC-013, TC-019 and TC-020 records and `perf-results/*.csv`; defects found during execution are listed in `docs/IMPLEMENTATION_STATUS.md` (all fixed and re-tested); test-reliability issues found in the re-verification of the same day (one intermittent TC-022 failure under load) are listed there; on 14-10-2026 the TC-022 failure was traced to the test harness, the changed test passed idle and once under CPU load, and a later loaded run (on battery power) finished only after the 300 s budget had elapsed (303.7 s; not a clean pass; TC-022 record). On 16-10-2026 the team decided to retain TC-022 as Pass with this timing overrun and the unresolved timing margin under load recorded as limitations (TC-022 record; `docs/reviews/TEAM_REVIEW_AGENDA.md` D-03 to D-07). On 17-10-2026 TC-021 was executed manually and, after a two-member review, set to Pass with its limitations recorded (TC-021 record); current totals: 34 test cases, 33 Pass, 0 Fail, 1 Blocked (TC-109); 49 of 50 requirements with RTM status P. The exit criteria of §5 are **not** yet met: TC-109 is Blocked without an approved exception, and the test summary has not been reviewed by all four members.

Regression run after the frontend redesign, 16-10-2026 (separate from the results above, which were obtained before it; automated only): the client's board view was restyled (layout, toolbar, side panel and dialog styling in `client/src`; no change to the server, the shared module, the APIs or the tests). On the same uncommitted working tree (base commit 82af885) afterwards: `npm run lint` clean; `npm run test:unit` 33/33 passed; `npm run test:integration` 53/53 passed; `npm run test:e2e` 10/10 passed in Playwright's Chromium; the same E2E suite with `E2E_BROWSERS=msedge,firefox` 19 passed and 1

skipped (TC-009 in Playwright's patched Firefox build, skipped by design) in Edge 155.0.4283.45 and Playwright's Firefox. Stable Firefox 157 and installed Chrome were not rerun, and no manual check was made, so TC-021 and the status of every test case are unchanged; TC-022 was not rerun under load.

Final regression on `main` after PR #33, 02-11-2026 (automated, no status change; full record `docs/reports/RELEASE_READINESS_2026-11-02.md`, logs `docs/evidence/2026-11-02-final-validation/`): lint passed; 22 Jest suites, 210 unit and integration tests passed; production build passed with no test hooks; 31 E2E tests passed in Chromium, installed Chrome and installed Edge, 28 passed and 3 skipped in Playwright's Firefox, 29 passed and 2 skipped in Playwright's WebKit (skips documented in the tests); traceability 1,483 checks passed; GitHub CI (`app-ci`, `docs-check`) passed on PR #33 and on `main`. Defects found and fixed in this cycle: top-bar layout below 1024 px (#23), unreadable login/register/join errors (#21), restore without `crypto.randomUUID()`, forced-colours swatches and lost pointer capture (#25); runtime dependency advisories fixed (`npm audit --omit=dev`: 0). Requirement coverage is unchanged: 49 of 50 with RTM status P; TC-109 Blocked.

15. Approvals

Role	Name	Signature / Date
QA Lead	Pranav V Menon	
Security Test Lead	Prithvi W	
Dev Lead (SAD)	P Chinni Krishna Reddy	
Product Owner (SRS)	Rachit S Mane	

Appendix A — Test case specifications

Execution 12-10-2026 (uncommitted working tree on base commit 82af885): 32 test cases Pass, 2 Blocked (TC-021: stable Firefox not installed at that time, still Blocked after the 16-10-2026 run in stable Firefox 157 because the manual checklist and the TC-009 offline check remain open; TC-109: no deployment), none Fail. TC-021 was executed manually on 17-10-2026 and set to Pass with limitations by team decision (TC-021 record), so 33 test cases are Pass and 1 Blocked (TC-109); details in each test case and in `docs/IMPLEMENTATION_STATUS.md`. The final regression of 02-11-2026 (§14) changed no status. "Actual result" records the evidence (test file or harness, measured values, build and date); allowed status values and their RTM mapping are defined in §13. The automated tests are in `server/tests/`, `shared/tests/`, `client/tests/` and `e2e/`; the performance harness and the TB-STD seed are in `scripts/perf/` (§6). Unless stated otherwise, every test case assumes: test environment of §6 running, health check ok, accounts seeded by the seed script, supported browser, test build with the hooks of SAD §4.1.6. Names used: O = Owner, E/E2 = Editors, V = Viewer, G = guest, M = other member.

A.1 Functional and non-functional test cases

TC-001 — Freehand stroke with colour and width is synchronised

Field	Value
Requirements	FR-001, FR-002, FR-020
Type / Priority	Functional, system (E2E), positive + boundary / High
Preconditions	O and E have board B open in two browsers.
Test data	Colour `#D32F2F`, stroke width 8; boundary widths 1, 20 (valid) and 0, 21 (invalid).
Steps	1. E selects colour `#D32F2F` and width 8. 2. E draws a stroke across the canvas. 3. Compare the stroke on O's board with E's (`exportState`). 4. E sets widths 1 and 20 and draws one stroke with each. 5. E tries to enter widths 0 and 21.
Expected result	Step 3: identical stroke on both boards, colour `#D32F2F`, width 8, path following the drag. Step 4: both strokes created with widths 1 and 20. Step 5: widths 0 and 21 are not accepted by the control.
Pass/fail criteria	Pass if every expected result holds; fail otherwise.
Dependencies / limitations	Requires FR-020 synchronisation; timing is not measured here (see TC-019).
Actual result	e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): stroke #D32F2F width 8 identical on both boards (exportState); widths 1 and 20 accepted; 0 and 21 marked invalid and not applied. Server side: server/tests/integration/sync.test.js. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-002 — Shapes and text are created and synchronised

Field	Value
Requirements	FR-003, FR-004, FR-020
Type / Priority	Functional, system (E2E), positive + boundary / High
Preconditions	O and E have board B open.
Test data	Text "Sprint plan"; strings of 500 and 501 characters.
Steps	1. E drags with the rectangle, ellipse and line tools. 2. E adds the text "Sprint plan", then edits it to "Sprint plan v2". 3. E enters a 500-character text, then tries a 501-character text. 4. Compare O's board with E's after each step.
Expected result	Each shape's bounding box or end points equal the drag start and end; the text appears and is updated on both boards; 500 characters are accepted and the 501st is not.
Pass/fail criteria	Pass if both boards are identical after every step and the length limit holds.
Dependencies / limitations	—
Actual result	e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): rectangle, ellipse and line bounding boxes/end points equal the drags (±1 px); text added, edited to 'Sprint plan v2' and synchronised; 500 typed characters accepted, the 501st not; both boards identical after every step. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-003 — Select, move, restyle and delete an object

Field	Value
Requirements	FR-005
Type / Priority	Functional, system (E2E) / High
Preconditions	Board B contains rectangle R; O and E connected.
Test data	New colour `#43A047`, new width 4, move of 100 px to the right.
Steps	1. E selects R and drags it 100 px to the right. 2. E changes R's colour to `#43A047` and width to 4. 3. E deletes R. 4. Compare O's board with E's after each step.
Expected result	O's board shows R moved, then restyled, then removed, matching E's board each time.
Pass/fail criteria	Pass if the boards match after every step.
Dependencies / limitations	—
Actual result	e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): R moved 100 px, restyled to #43A047 width 4, deleted; O's board matched E's after each step. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-004 — Eraser removes whole touched objects

Field	Value
Requirements	FR-006
Type / Priority	Functional, system (E2E) / High
Preconditions	Board B contains strokes S1, S2 and text T; S1 and T lie on the planned eraser path, S2 does not.
Test data	—
Steps	1. E selects the eraser and drags across S1 and T. 2. Compare both boards.
Expected result	S1 and T are removed completely; S2 is unchanged.
Pass/fail criteria	Pass if exactly S1 and T are deleted on both boards.
Dependencies / limitations	—
Actual result	e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): eraser path across S1 and T removed exactly S1 and T on both boards; S2 unchanged. Hit testing: client/tests/geometry.test.js. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-005 — Clearing a board and the "before clear" version

Field	Value
Requirements	FR-007, FR-030
Type / Priority	Functional, positive + negative + boundary / High
Preconditions	Board B with at least 5 objects; O, E and V connected; latest version number L recorded.
Test data	—
Steps	1. E submits `board:clear` with a scripted client using E's token. 2. O chooses Clear board and cancels the confirmation. 3. O makes one change (board now has changed content), chooses Clear board and confirms. 4. Open the version list; preview the newest version. 5. O adds 3 objects, waits until a periodic version is created (no further change), then clears and confirms. 6. Open the version list again. 7. O clears the now empty board and confirms.
Expected result	Step 1: rejected with `FORBIDDEN`; board unchanged. Step 2: nothing changes. Step 3: board empty for O, E and V. Step 4: a new "before clear" version L+1 exists and contains the pre-clear objects. Step 6: no additional "before clear" version (no-change rule); the newest version (the periodic one) contains the 3 objects. Step 7: no change, no new version, acknowledgement `noop: true`.
Pass/fail criteria	Pass if every expected result holds.
Dependencies / limitations	Step 5 waits for interval *T* of the build under test (record the configured value; the production value is 5 min, D11a). Rejection of a clear whose before-clear version cannot be stored (FR-007) is tested in TC-025 (step 1, fault injection), not in this system test.
Actual result	server/tests/integration/sync.test.js: Editor clear FORBIDDEN; before-clear version holds the pre-clear objects; no extra version after a periodic version (periodic trigger invoked directly instead of waiting T); empty clear noop:true without version. E2E: cancel changes nothing; confirmed clear empties both boards. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-006 — Concurrent edits to different objects

Field	Value
Requirements	FR-021, FR-022
Type / Priority	Functional, integration (scripted clients), concurrency / High
Preconditions	Board B contains objects X and Y; scripted clients C1 and C2 joined with Editor tokens, in one process.
Test data	C1 moves X; C2 recolours Y; 50 rounds.
Steps	1. In each round, C1 and C2 submit their updates at the same moment (shared start signal). 2. After each round, compare `{lastSeq, objects}` of C1, C2 and a fresh `board:join`. 3. Inspect the sequence numbers each client applied.
Expected result	In every round both updates are present on all three states; on every client the applied sequence numbers are strictly increasing, without gaps or duplicates, and in the same order.
Pass/fail criteria	Pass if 50/50 rounds converge with both changes present.
Dependencies / limitations	—
Actual result	server/tests/integration/sync.test.js: 50/50 rounds converged (both clients and a fresh board:join identical); applied seq strictly increasing, contiguous and in the same order. Rounds paced ≥ 50 ms (30 ops/s limit). Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-007 — Concurrent edits to the same object (last write wins)

Field	Value
Requirements	FR-021, FR-023
Type / Priority	Functional, integration (scripted clients), concurrency / High
Preconditions	Board B contains object Z; scripted clients C1, C2 (Editors) and C3 (observer) joined, in one process.
Test data	Same property: C1 sets colour `#1E88E5`, C2 sets colour `#FB8C00`. Different properties: C1 moves Z, C2 sets width 6.
Steps	1. C1 and C2 update Z's colour at the same moment; record both sequence numbers; repeat 50 times. 2. Compare Z on C1, C2 and C3 after each round. 3. C1 moves Z while C2 changes its width at the same moment; compare. 4. C1 deletes Z while C2 updates Z at the same moment; record acknowledgements.
Expected result	Step 2: in every round all clients show the colour of the update with the higher sequence number. Step 3: Z has both the new position and width 6 on all clients. Step 4: if the delete has the lower sequence number, the update is rejected with `OBJECT_NOT_FOUND` and Z stays deleted everywhere; otherwise both are accepted and Z is deleted.
Pass/fail criteria	Pass if all rounds converge as expected.
Dependencies / limitations	—
Actual result	server/tests/integration/sync.test.js: 50 rounds, all clients show the colour of the higher seq; move + width 6 merged; delete vs update: update rejected OBJECT_NOT_FOUND or ordered before the delete, Z deleted everywhere. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-008 — Late joiner receives full state

Field	Value
Requirements	FR-024
Type / Priority	Functional, system / High
Preconditions	O and E have made at least 20 changes to board B.
Test data	—
Steps	1. E2 opens board B. 2. Compare E2's `exportState()` with O's. 3. Check E2's participant list.
Expected result	Identical `lastSeq` and objects; E2 sees O and E in the participant list.
Pass/fail criteria	Pass if states are identical and the list is correct.
Dependencies / limitations	—
Actual result	server/tests/integration/sync.test.js: late joiner after 20 changes has identical lastSeq and objects; participant list shows Owner, Editor, Viewer. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-009 — Reconnection recovery

Field	Value
Requirements	FR-025, NFR-004
Type / Priority	Functional + reliability (recovery), system / High
Preconditions	Board B contains TB-STD; O (scripted client) and E (browser, Playwright) connected.
Test data	20 changes by O during the outage; one unacknowledged change by E (see step 2).
Steps	1. Set E's browser context offline (`setOffline(true)`) for 30 s. 2. During the outage O makes 20 changes; E attempts to draw (tools should be disabled). In one repetition, make E draw a stroke and go offline before its acknowledgement arrives (by blocking the WebSocket frame). 3. Call `setOffline(false)` and record t0 (test clock). 4. Record t1 = time of E's `board-ready` mark after resynchronisation; recovery = t1 − t0. 5. Compare E's `exportState()` with a fresh `board:join`. 6. Repeat 10 times; record every value.
Expected result	During the outage the indicator is shown and drawing is impossible. After reconnection E's state equals the server state including O's 20 changes; in the repetition with an unacknowledged stroke, that stroke is gone unless the server had stored it, and the "unconfirmed changes" notice is shown; recovery ≤ 5 s.
Pass/fail criteria	Pass if all 10 repetitions match the server state and each recovery ≤ 5 s (NFR-004); the measured values are reported in any case.
Dependencies / limitations	Depends on the reconnection settings of SAD §4.1.4; times are from one clock in the Playwright process.
Actual result	Banner and disabled tools during the outage, resync equal to server state: e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge). 10 repetitions, 30 s outage, 20 changes (scripts/perf/browser.js): recovery max 2194 ms (Chromium), max 2322 ms (chrome 154.0.8037.99); state equal every time. Unacknowledged-change repetition (acknowledgement dropped, connection cut): notice shown, state equals server (E2E). Not simulable in Playwright's Firefox (offline emulation leaves WebSockets open). Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-010 — Rejected operation is rolled back

Field	Value
Requirements	FR-026
Type / Priority	Functional, negative, system / High
Preconditions	O and E have board B open; E's test build exposes `submitRaw`.
Test data	Rectangle with `strokeWidth` 50 (invalid).
Steps	1. E submits the invalid rectangle with `submitRaw` (it is rendered optimistically). 2. Record the time the rejection arrives and the time the rectangle disappears from E's view. 3. Check E's message and O's board.
Expected result	Server rejects with `VALIDATION_FAILED`; E sees an error message; the rectangle disappears within 1 s of the rejection; O never receives it.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Uses a test-build hook; not possible in a production build.
Actual result	e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): submitRaw strokeWidth 50 → VALIDATION_FAILED message, rectangle removed (< 1 s, sync.test.js), O never received it. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-011 — Board survives server restart

Field	Value
Requirements	FR-027
Type / Priority	Functional, recovery / High
Preconditions	Board B open by O (scripted client).
Test data	30 operations, each acknowledged.
Steps	1. O submits 30 changes and waits for all 30 acknowledgements; record O's `{lastSeq, objects}`. 2. All participants leave. 3. Stop the server process (kill) and restart it. 4. O rejoins board B and compares the state with step 1.
Expected result	The rejoined state contains all 30 acknowledged changes and `seq` continues above the last acknowledged value.
Pass/fail criteria	Pass if no acknowledged change is missing.
Dependencies / limitations	MongoDB must use journaling (default).
Actual result	server/tests/integration/recovery.test.js: 30 acknowledged changes, server process killed and restarted; rejoined state identical; next seq 31. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-012 — Automatic versions, no-change rule, list and preview

Field	Value
Requirements	FR-030, FR-031, FR-032
Type / Priority	Functional, system, positive + no-change / High
Preconditions	Board B open by O, E and V; interval *T* as configured in the build under test (record it; a test build may set e.g. 30 s via `VERSION_INTERVAL_MS`). Record latest version number L.
Test data	—
Steps	1. E makes a change and waits one interval *T*. 2. Without further changes, wait two more intervals. 3. E makes another change; all participants leave; O reopens B. 4. All participants leave again without changes; O reopens B. 5. V opens the version list. 6. V previews version L+1; during the preview, E makes a change.
Expected result	Step 1: exactly one "periodic" version L+1. Step 2: no new version. Step 3: exactly one "last participant left" version L+2. Step 4: no new version. Step 5: list shows L+2, L+1, … newest first with number, time and trigger. Step 6: preview shows version L+1's objects read-only; E's change appears on the live boards and not in the preview.
Pass/fail criteria	Pass if version counts, triggers, list fields and preview behaviour all match.
Dependencies / limitations	Result is valid for the interval configured in the build under test; the production value is 5 min (D11a). Steps 3 and 4 must each be completed within one interval so that no periodic trigger interferes.
Actual result	server/tests/integration/versions.test.js: VERSION_INTERVAL_MS=1000: one periodic version, none in two unchanged intervals; last_left version once, none on a no-change leave (timer disabled for these steps); list newest first with number, createdAt, trigger; preview unchanged by a live edit. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-013 — Owner restore creates a new version for all participants

Field	Value
Requirements	FR-033, FR-034, NFR-005
Type / Priority	Functional + performance, system / High
Preconditions	Board B with TB-STD content saved as version n; afterwards content changed and saved as later versions; 10 scripted clients connected in one process (one of them O); one browser client for the UI check.
Test data	TB-STD.
Steps	1. UI check: O (browser) selects version n, confirms Restore; check the browser board and version list. 2. Timing run: the scripted O sends `POST …/versions/n/restore` with a new `Idempotency-Key` for every repetition (each repetition is a distinct restore); record t0 when the request is sent. 3. Record, for each of the 10 clients, the time it applies `board:state`; t1 = the latest of these. 4. Repeat steps 2–3 ten times; restore = t1 − t0 each time. 5. Restore the version that is currently identical to the board content once more.
Expected result	All participants show version n's content without reload; the list contains every previously retained version unchanged plus a new "restore of n" version; restore ≤ 3 s; step 5 also creates a new "restore" version.
Pass/fail criteria	Pass if content and history are correct in all runs and 10/10 timings ≤ 3 s (NFR-005); measured values are reported in any case.
Dependencies / limitations	Timing uses one clock in the harness process; browser rendering of the restored board is checked in step 1 but not timed.
Actual result	UI restore (E2E) and history unchanged plus 'restore of n' (versions.test.js); identical-content restore creates a version. Timing, TB-STD, 10 clients, 10 runs (scripts/perf/restore.js): max 1661 ms, median 1005.1 ms (≤ 3,000 ms). Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-014 — Version retention

Field	Value
Requirements	FR-035
Type / Priority	Functional, integration, boundary / Medium
Preconditions	New board B; short `VERSION_INTERVAL_MS` configured.
Test data	25 versions with distinct content (one change before each interval).
Steps	1. Create 25 versions. 2. List versions. 3. Preview and restore the oldest listed version. 4. Request the preview of version 1 by number.
Expected result	At least the 20 newest versions are listed; the oldest listed one can be previewed and restored; if version 1 is no longer retained, the request returns `404 VERSION_NOT_FOUND`.
Pass/fail criteria	Pass if ≥ 20 newest versions are available and usable and removed versions return `VERSION_NOT_FOUND`.
Dependencies / limitations	Retention count 20 (D5a).
Actual result	server/tests/integration/versions.test.js: 25 versions → 20 newest listed; version 6 previewed and restored (new version 26); version 1 → 404 VERSION_NOT_FOUND. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-015 — Register, log in, log out

Field	Value
Requirements	FR-040, FR-041
Type / Priority	Functional, integration (API) + E2E, positive + negative + boundary / High
Preconditions	E-mail address `tc015@example.test` not registered.
Test data	Valid password generated at run time; display names of 0, 1, 30 and 31 characters.
Steps	1. Register with valid data (display name of 1 character), then a second account with a 30-character display name. 2. Register again with the first e-mail address. 3. Register with display names of 0 and 31 characters. 4. Log in with a wrong password, then with the correct one. 5. Log out and try to open the dashboard.
Expected result	Step 1 succeeds twice; step 2 → `EMAIL_TAKEN`; step 3 → `VALIDATION_FAILED`; wrong password → generic "invalid e-mail or password"; correct password opens the dashboard; after logout the dashboard redirects to login.
Pass/fail criteria	Pass if every step gives the expected outcome.
Dependencies / limitations	Password rules are tested in TC-105.
Actual result	server/tests/integration/auth.test.js (display names 1/30 accepted, 0/31 rejected, EMAIL_TAKEN, generic login failure, logout revokes) and E2E register → dashboard → logout → login page. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-016 — Create, list and delete boards

Field	Value
Requirements	FR-042, FR-043, FR-044
Type / Priority	Functional, system, positive + boundary / High
Preconditions	Registered users O and M; M is an Editor member of O's existing board B1 and is connected to it.
Test data	Titles of 0, 1, 100 and 101 characters.
Steps	1. O creates board B2 with a 100-character title and B3 with a 1-character title; try 0 and 101 characters. 2. Check O's and M's dashboards. 3. O deletes B1 and confirms. 4. Open one of B1's share links.
Expected result	B2 and B3 created with role Owner, invalid titles rejected; O's dashboard lists B1–B3, M's lists B1 as Editor; after deletion M sees "board deleted" and is disconnected, B1 disappears from both dashboards, and the link is rejected.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Non-Owner deletion is tested in TC-102.
Actual result	server/tests/integration/access.test.js (titles 1/100 accepted, 0/101 rejected; dashboards; deletion cascade; link rejected) and E2E dashboard delete with 'board deleted' notice at the connected member. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-017 — Share link creation, joining and presence

Field	Value
Requirements	FR-045, FR-046, FR-052
Type / Priority	Functional, system / High
Preconditions	O has board B open; registered user R (not a member) is logged in; a third browser is not logged in.
Test data	Editor link (24 h); Viewer link (default expiry); guest name "Asha"; guest names of 0 and 31 characters.
Steps	1. O creates an Editor link with 24 h expiry and a Viewer link with the default expiry. 2. Check O's link list and that the full URL is shown only at creation. 3. R opens the Editor link. 4. The third browser opens the Viewer link; try names of 0 and 31 characters, then join as "Asha". 5. Asha leaves.
Expected result	Both links listed with correct role and expiry (default 7 days); R becomes an Editor member and can draw; invalid guest names rejected; "Asha (guest)" appears in every participant list with Viewer rights and no drawing tools; after Asha leaves, all lists update.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Revocation and expiry are tested in TC-103.
Actual result	access.test.js (link list, expiry 24 h / default 7 d, token shown once, member upgrade, guest names) and E2E: link created in the Share dialog, guest 'Asha (guest)' in participant list with View only, registered user joins as Editor after logging in. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-018 — Ownership transfer

Field	Value
Requirements	FR-050
Type / Priority	Functional, integration (API) + system, positive + negative / High
Preconditions	Board B with Owner O, Editor member E, Viewer member V, guest G and a registered non-member X; O and E connected in browsers.
Test data	Endpoint `POST /api/v1/boards/{B}/transfer-ownership`.
Steps	1. E calls the endpoint with `{newOwnerId: V, confirm: true}`. 2. G (guest token) calls it with `{newOwnerId: E, confirm: true}`. 3. O calls it with `{newOwnerId: E}` (no `confirm`). 4. O calls it with `newOwnerId` = O, = X, = G's guest ID and = an unknown ID (each with `confirm: true`). 5. O calls it for a non-existent board ID. 6. O uses the Members dialog to transfer ownership to E and confirms. 7. Check E's and O's tools without reload, both dashboards and the members list. 8. O tries to create a link, restore a version, change V's role and transfer ownership back through the API, and sends `board:clear` over the socket that was opened while O was Owner. 9. E (new Owner) creates a share link, restores a version and changes V's role to Editor. 10. E deletes board B and confirms.
Expected result	Step 1: 403 `FORBIDDEN`. Step 2: 403 `FORBIDDEN`. Step 3: 400 `VALIDATION_FAILED`. Step 4: 400 `INVALID_TARGET` each time. Step 5: 404 `BOARD_NOT_FOUND`. Steps 1–5 change no role. Step 6: 200 with `ownerId` = E and `previousOwner.role` = `editor`. Step 7: E sees Owner controls and O sees Editor tools without reload; dashboards show E as Owner and O as Editor; E no longer appears as a member row. Step 8: every API action of O is rejected with 403 and the `board:clear` is rejected with `FORBIDDEN` (no stale Owner permission on the open socket). Steps 9–10: every action of E succeeds and the board is deleted.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Unauthenticated calls are covered by TC-106.
Actual result	access.test.js steps 1–10 (403, 403, 400, 400 INVALID_TARGET ×4, 404, roles unchanged, 200, former Owner rejected incl. board:clear on the old socket) and E2E transfer with confirmation, roles updated without reload. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-019 — Synchronisation latency and capacity

Field	Value
Requirements	NFR-001, NFR-002
Type / Priority	Performance / High
Preconditions	Board B empty; latency harness with 10 scripted Editor clients in one Node.js process using `performance.now()`; environment conditions of §6 and SRS §5.
Test data	Each client submits 2 valid operations per second (random adds and updates of its own objects, well below the 30/s rate limit).
Steps	1. Connect all 10 clients; run 10 s of warm-up (not recorded). 2. For 60 s, for every operation record t_submit at the sender immediately before `emit("op:submit")`, the acknowledgement time at the sender, and t_apply at each of the 9 receivers when the operation has been applied to its state. 3. Count rejected operations; they are excluded from latency samples and reported separately. 4. Latency samples = t_apply − t_submit for every (accepted operation, receiver) pair (≈ 10,800); compute p95 by the nearest-rank method; also report median, p99 and maximum, and the sender acknowledgement times separately. 5. Check every client's applied sequence numbers for missing or duplicate values and compare all final states with a fresh `board:join`. 6. Record results in a CSV with commit hash, hardware and software versions.
Expected result	p95 ≤ 500 ms; no disconnection; zero rejected operations; no missing or duplicate sequence numbers; identical final states.
Pass/fail criteria	Pass if all conditions hold against the NFR-001 and NFR-002 targets; measured values are reported in any case.
Dependencies / limitations	Rendering time in browsers is not included (scripted clients apply but do not render). Results are valid only for the recorded hardware.
Actual result	scripts/perf/latency.js, 10 clients, 10 s warm-up, 60 s: p95 80 ms, median 34.7, p99 181.4, max 383.2 ms, 10566 samples; rejected 0, disconnects 0, seq problems 0, final states identical. Results in perf-results/tc019-latency.csv. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-020 — Initial board load time

Field	Value
Requirements	NFR-003
Type / Priority	Performance / Medium
Preconditions	Board B contains TB-STD; logged-in Owner session available to Playwright.
Test data	TB-STD.
Steps	1. Create a new browser context (empty cache) and navigate to board B. 2. Read the `board-ready` mark's `startTime` (ms since navigation start). 3. Repeat 10 times; record every value and the browser version.
Expected result	Every run ≤ 3,000 ms.
Pass/fail criteria	Pass if 10/10 runs ≤ 3 s (NFR-003); all values reported in any case.
Dependencies / limitations	Run in Chrome; other browsers are covered functionally by TC-021.
Actual result	scripts/perf/browser.js, TB-STD, new context per run, 10 runs: max 1110 ms in chrome 154.0.8037.99; max 380.6 ms in Playwright Chromium (≤ 3,000 ms). Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-021 — Browser compatibility, viewport and usability checklist

Field	Value
Requirements	NFR-006, NFR-007
Type / Priority	Compatibility + usability, manual / Medium
Preconditions	Current stable Chrome, Firefox and Edge installed; versions recorded.
Test data	Viewport 1280 × 720.
Steps	1. In each browser at 1280 × 720, run the core flows of TC-001, TC-002, TC-013 (step 1) and TC-017. 2. Check that no control is horizontally cut off. 3. For each role (O, E, V), check: role badge visible; connection status visible; all drawing tools directly on the toolbar; every toolbar control has visible text or a tooltip; every toolbar control can be reached with Tab and activated with Enter or Space.
Expected result	Core flows succeed in all three browsers; no cut-off controls; every checklist item passes for every role.
Pass/fail criteria	Pass if all flows and all checklist items pass.
Dependencies / limitations	Manual inspection; not a formal accessibility audit.

Field	Value
Actual result	Stable Mozilla Firefox is not installed on the test machine. Chrome 154 and Edge 155 (installed stable channels): every E2E core flow (TC-001, TC-002, TC-013, TC-017) and the automated checklist e2e/usability.spec.cjs (role badge, connection status, all tools on the toolbar, labelled controls, Tab reach, Enter/Space activation, no cut-off at 1280 px) passed for Owner, Editor and Viewer. Playwright's patched Firefox 134 build passed the same, except TC-009 (offline emulation not possible there); it does not count as stable Firefox. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1. 14/15-10-2026 addendum (automated only; same working tree): the E2E suite including `e2e/usability.spec.cjs` passed again in Chrome 155.0.8059.41 and Edge 155.0.4283.45 and in Playwright's Firefox build (29 passed, 1 skipped — TC-009 in that build), in four runs (14-10-2026 twice, 15-10-2026 twice). Reference screenshots of the Owner, Editor and Viewer board views at 1280 × 720 were captured in all three browsers; they show the role badge ("Owner", "Editor", "View only") and "Connected" as text, all seven tools on the toolbar for Owner and Editor, and no drawing tools for the Viewer. Stable Mozilla Firefox is still not installed (re-checked 15-10-2026), and no manual run or human usability judgement has been recorded in any browser. 16-10-2026 addendum (automated only; same working tree): stable Mozilla Firefox 157.0.1 (release channel; official Mozilla installer, signature valid) was extracted with the installer's `/ExtractDir` option and run without installation, because installation needs administrator approval. Driven through Playwright 1.50's experimental WebDriver BiDi driver: 8 passed, 2 failed. Both failures are driver limitations, shown by a probe without application code: the driver sends two single clicks and no `dblclick` event (TC-001 to TC-004 text edit), and `setOffline()` leaves `navigator.onLine` true, so no outage occurs (TC-009). TC-001 to TC-004 then passed in full with the double-click dispatched as a DOM event; the automated checklist `e2e/usability.spec.cjs` passed; Owner, Editor and Viewer screenshots at 1280 × 720 were captured. TC-009 is not verified in stable Firefox. The manual checklist and human judgement are still outstanding. 17-10-2026 manual execution and team decision (record: `docs/evidence/2026-10-17-tc021-firefox/RESULTS.md`; uncommitted working tree on base commit 82af885; local app via `npm run dev`). Manual, tester-reported, in installed Mozilla Firefox 158.0 (64-bit) at 1280 × 720 (Responsive Design Mode, DPR 1): the TC-001, TC-002, TC-013 step 1 and TC-017 flows passed; for Owner, Editor and Viewer the role badge, the "Connected" text, the seven tools on the toolbar (none for the Viewer), no cut-off and no horizontal scroll passed; tooltips on the seven tools and the keyboard checks passed. Manual, tester-reported, in Chrome at 1280 × 720: layout, role badges, "Connected" text, the seven tools and their tooltips, Tab navigation, Enter/Space activation and colour-swatch selection by keyboard passed; the core flows were not run by hand in Chrome. Edge was not checked by hand (tester's decision). Automated, 17-10-2026 20:17 IST, Chrome 155.0.8059.82 and Edge 155.0.4283.45, on the client test build of 16-10-2026: `e2e/usability.spec.cjs` and the E2E tests for TC-001 to TC-004, TC-005/TC-013 and both TC-017 tests, 10 passed, 0 failed, 0 skipped. Screenshots: About Firefox, the 1280 × 720 viewport and the Owner, Editor and Viewer boards in Firefox. Defect found and fixed during TC-021: the stroke-width "Increase" and "Decrease" buttons had no visible text and no tooltip (only an `aria-label`, which the automated checklist accepts); the existing tooltip pattern was added to both in `client/src/board/Toolbar.jsx` (17-10-2026), `npm run lint` clean; the tester re-checked in Firefox only that both tooltips appear on hover and on Tab focus and disappear when the button is disabled at widths 20 and 1. Two team members reviewed the record and the team accepted these limitations: the exact name stored for the 31-character guest-name attempt was not recorded (the team accepts the 30-character input limit as "not accepted"); the manual TC-017 flow did not follow this test's role sequence exactly (a guest used the Editor link instead of a registered non-member); the tooltip fix was re-checked in Firefox only, with no post-fix Chrome/Edge check and no automated run after the fix; the Editor keyboard result was accepted without separate role-specific evidence; manual Edge testing was skipped, and the automated Chrome/Edge run predates the fix; tooltip and core-flow end-state screenshots and Chrome screenshots were not captured; the build is uncommitted. TC-009 is not part of TC-021.
Status	Pass

Addendum 02-11-2026 (status unchanged): the E2E suite (31 tests, including the TC-001, TC-002, TC-013 and TC-017 flows and `e2e/usability.spec.cjs`) passed at 1280 × 720 in installed Chrome and Edge and in Playwright's Firefox and WebKit builds (documented skips only); in installed stable Mozilla Firefox 159 through Playwright's experimental BiDi driver 22 tests passed and 7 failed, each traced to a driver limitation (`docs/evidence/2026-11-02-final-validation/22-stable-firefox-diagnosis.txt`). The responsive layout was additionally checked at 1366 × 768, 1920 × 1080 and narrower widths (`e2e/responsive.spec.cjs`). Apple Safari was not tested.

TC-022 — Board content size limit

Field	Value
Requirements	FR-008
Type / Priority	Functional, integration (scripted client + database inspection), boundary + negative / Medium
Preconditions	Empty board B; scripted Editor client E and observer client O (both in one Node.js process); a test server with the periodic timer disabled (`VERSION_INTERVAL_MS=0`); version creation is triggered through the same function the timer calls; read access to the test MongoDB.
Test data	Produced by the deterministic generator `gen-board-fill` (fixed seed): strokes of at most 2,000 points (the object schema limit, SAD §4.1.1) whose JSON-encoded event payload is at most 60,000 bytes, leaving at least 5,536 bytes for Socket.IO framing below the 65,536-byte message limit (M1). For every generated object the generator records the encoded packet size (M1), `BSON.calculateObjectSize` of the object in snapshot form (M2) and the running content size M3 = Σ(M2 + 9). No size is assumed from the point count.
Steps	1. Fill: E submits generated strokes one at a time (each submission waits for its acknowledgement and then 40 ms, i.e. fewer than 25 per second, below the 30 operations/s limit), until R = 15,728,640 − M3 is smaller than the M2 + 9 of a typical stroke. Record the largest M1 (every M1 is checked against 65,536 bytes) and the final M3; cross-check M3 with the board's stored content size (`boards.contentBytes`) and, for the rejected operation of step 3, with `attemptedBytes` in the `op.rejected` log line. 2. Boundary — fits: build an object (stroke or text, adjusting points or text length) with M2 + 9 ≤ R and as close to R as achievable; record its M2 + 9 and the resulting M3; submit it. 3. Boundary — exceeds: compute the new R; build an object with M2 + 9 = R + 1 byte (or the smallest achievable value above R; record it); submit it. 4. Inspect: compare `exportState()` of E and O and the `boardObjects` count and `boards.seq` in MongoDB before and after step 3; check the log for `op.rejected` with `BOARD_FULL`. With M3 at the limit, also submit an update that grows an object, a new object, and an update that does not grow the content. 5. Version near the limit: trigger a periodic version directly (same code path as the timer); measure the BSON size of the newest version document with MongoDB's `$bsonSize` (equivalent to `Object.bsonsize` in `mongosh`). 6. Recovery after deletion: delete two strokes, then submit a new stroke whose M2 + 9 is smaller than the space freed.
Expected result	Step 1: every message M1 ≤ 65,536 bytes; every operation accepted. Step 2: accepted; resulting M3 ≤ 15,728,640. Step 3: rejected with `BOARD_FULL`; E removes its optimistic stroke and shows the message (the scripted client checks the removal only, see Dependencies); O receives no `op:applied`; `boards.seq`, object count and content are identical before and after (step 4); at the limit the growing update and the new object are rejected with `BOARD_FULL` and the non-growing update is accepted. Step 5: a new version exists; its BSON size is ≤ 16,711,680 bytes (defensive threshold) and therefore below 16,777,216 bytes; the test also checks the tighter bound 15,728,640 + 286 + 1,024 bytes (D11c bound plus the version metadata fields). Step 6: accepted.

Field	Value
Pass/fail criteria	Pass if every expected result holds, using the recorded M1, M2, M3 and M4 values as evidence (all values are reported). Fail if any message exceeds 65,536 bytes, any over-limit operation is stored or broadcast, or the version is missing or above the threshold.
Dependencies / limitations	The generator (`gen-board-fill` in `server/tests/integration/boardLimit.test.js`), the scripted clients and the test hooks are implemented (historical v1.0 note: they were planned and not yet implemented). The 15 MiB limit and its accounting are specified in D11c. The M4 margin is derived in SAD §3.6; this test supplies the measurement that confirms it. Rejection of a clear whose version cannot be stored and recovery of an interrupted clear are covered by TC-025. TC-022 checks the rejection through the client sync module only (scripted clients, no UI): whether the "Board is full" message appears in the browser is not covered by its evidence. The 300 s limit is the test's Jest time budget (`jest.setTimeout(300000)`), not a pass/fail criterion of this test case.

Actual result	server/tests/integration/boardLimit.test.js: 243 generated strokes; max M1 31,881 bytes; object of exactly R + 1 bytes rejected BOARD_FULL with no change or broadcast; object of exactly R accepted (M3 = 15,728,640); version M4 15,727,703 bytes; after deleting two strokes a new stroke accepted. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1. Re-verification the same day (`docs/evidence/2026-10-12-reverification/`), five full `npm test` runs: run 1 — TC-022 passed (another suite failed, see the status report); run 2, on a heavily loaded machine — TC-022 failed at step 6: deleting stroke `fill-0000` was rejected; the test does not record the error code and the cause is not known; runs 3, 4 and 5 — all 77 tests passed. TC-022 run alone afterwards: passed with the values above. Kept as Pass on Rachit's decision of 12-10-2026 with this reliability issue open and team review pending (`docs/IMPLEMENTATION_STATUS.md` §6). 14-10-2026 addendum: the step-6 failure was reproduced under artificial CPU load (16 busy processes on 8 logical CPUs) with the error code now recorded — `NOT_READY`: both test clients had been disconnected by the Socket.IO heartbeat (`ping timeout`) and the reconnecting client did not send the delete, so the server never received it (`docs/evidence/2026-10-14-tc022-investigation/`). The test was changed: it records the error code, disconnects and timings, and before step 6 waits (at most 60 s) until both clients have joined again; steps 1–5 and all expected results are unchanged. Final test version, same day, uncommitted working tree (base commit 82af885), Windows 10, Node.js 26.7.0, MongoDB 8.2.1: passed alone and in two full `npm test` runs on an idle machine, and passed once under the same 16-process load (Jest time 295.6 s of the 300 s timeout; two `ping timeout` disconnects at 209.6 s, both clients rejoined; max M1 31,881 bytes; M4 15,727,703 bytes; `docs/evidence/2026-10-14-full-verification/15-tc022-cpuload-final.txt`). Why heartbeats are missed under load is not explained. Status unchanged; the team review of this record is still pending (`docs/reviews/TEAM_REVIEW_AGENDA.md` D-03 to D-07). Later on 14-10-2026 (`docs/evidence/2026-10-14-tc022-performance/`): the heartbeat misses were traced to multi-second event-loop stalls in the test process, which hosts the server and both clients (11.1 s and 6.1 s stalls during step 5 on an idle machine running on battery); the earlier statement that the event loop never stalled for more than 133 ms came from a step-5-only measurement that cannot record a stall starting before its first sample, and is withdrawn. Inside Jest's module sandbox the BSON work of the 15 MiB version runs about 15 times slower than in plain Node.js (measured: 4,957 ms versus 324 ms for the size calculation). Two changes followed: the server now writes object additions and updates through the native driver after schema validation (Mongoose re-casting of 2,000-point strokes cost about ten times the write; stored documents verified identical by `server/tests/integration/persistence.test.js`), and the test measures M4 with MongoDB's `$bsonSize` instead of reading the 15 MiB document back (same value, 15,727,703 bytes). Idle, on battery: 147 s before, 67.4 s after, no disconnects. Under the 16-process load, on battery: passed, but the test's own timer measured 303.7 s from start to end against the 300 s timeout (one 52.7 s stall during the version step; two `ping timeout` disconnects, both clients rejoined), so this is not a clean pass; options are listed under D-07. 15-10-2026 addendum (same uncommitted working tree, Windows 10, Node.js 26.7.0, MongoDB 8.2.1, **AC power**; logs kept outside the repository pending team review): TC-022 alone, three runs back to back on an idle machine — all passed, Jest 39.0 s, 38.6 s and 38.6 s (32.7 s measured inside the test each time), acknowledgement p50 29–30 ms, no disconnects, one event-loop stall of 4.1–4.2 s per run during the version step (step 5); under the same 16-process load — passed, Jest 174.6 s (143.4 s inside the test) against 295.6 s for the loaded run before the persistence change (also on AC power), with a 36.2 s stall during the version step and two `ping timeout` disconnects at 102.7 s, both clients rejoined. Also `npm test`: 14 suites, 85 passed. The remaining stall is the synchronous work of the 15 MiB version snapshot (`Version.create` through Mongoose and the BSON size check) in the process that also hosts both test clients. Status unchanged; D-03 to D-07 still pending. Later on 15-10-2026, same code, **on battery**: TC-022 alone twice — passed, Jest 101.2 s (a 12.3 s stall in step 5 and two `ping timeout` disconnects, both clients rejoined, **without artificial load**) and 58.6 s (6.4 s stall, no disconnect); under the 16-process load — passed, Jest 290.6 s (260.7 s inside the test, 9.4 s below the 300 s timeout), 84.5 s stall in step 5, two disconnects, both rejoined; `npm test` 14 suites, 86 passed. A measurement of the version write alone (full 15 MiB board, three repetitions) found the event loop blocked 178–277 ms by the size check and 354–536 ms by

Field	Value
	`Version.create` in plain Node.js, and 2.2–2.8 s and 2.4–4.0 s respectively inside Jest; writing the version through the native driver instead of Mongoose blocked 272–433 ms (plain) and 1.9–3.2 s (Jest), so the stall is dominated by BSON processing of the snapshot, which Jest's module sandbox slows by roughly ten times. Status unchanged. **Team decision, 16-10-2026** (`docs/reviews/TEAM_REVIEW_AGENDA.md` §4, D-03 to D-07, as communicated to the documentation session; individual approvals and meeting minutes are not recorded): the status is retained as **Pass**, with the following evidence acknowledged. (1) Time budget: the test runs under a 300 s Jest timeout, a test-harness budget and not a pass/fail criterion of TC-022. (2) The loaded run of 14-10-2026 (16-process CPU load, on battery; `docs/evidence/2026-10-14-tc022-performance/10-cpuload-after.txt`) took 303.7 s from test start to end by the test's own timer: it did **not** meet the 300 s budget and is **not** a clean pass, although Jest reported it as passed and all its assertions held. The recorded explanation, not independently verified, is that Jest's timeout timer was itself delayed by the 52.7 s event-loop stall in the version step. (3) Other recorded runs, passed unless stated: 12-10-2026 — alone, and runs 1, 3, 4 and 5 of the re-verification; run 2 (heavily loaded) failed at step 6. 14-10-2026 — changed test alone and in two full `npm test` runs (idle), one 16-process loaded run in 295.6 s; after the persistence change, idle on battery in 67.4 s. 15-10-2026 — idle 39.0, 38.6 and 38.6 s and loaded 174.6 s (AC power); idle 101.2 and 58.6 s and loaded 290.6 s (battery). 16-10-2026 — idle, within the integration suite after the frontend redesign (STP §14). (4) Explanation: the step-6 failure mechanism (both scripted clients dropped by the Socket.IO heartbeat, `NOT_READY`, the delete never sent) was reproduced under load on 14-10-2026; that run 2 of 12-10-2026 had the same cause is consistent with this but not confirmed. The stalls are attributed to BSON processing of the 15 MiB version snapshot inside Jest's module sandbox, where the server and both clients share one process (measured roughly 10–15 times slower than plain Node.js); that the timing overrun is a harness effect rather than an application fault is the team's working explanation and is not proven by TC-022. (5) Rationale (`docs/reviews/TC022_CORRECTION_PROPOSAL.md`, items D and E): apart from run 2 of 12-10-2026, every expected result the test checks held in every run; no recorded TC-022 value changed; the failure and the overrun concern the test harness rather than a violated FR-008 criterion. Limitations accepted with the decision: the timing margin under load is not resolved, no loaded run was made after 15-10-2026, and the test has not been run on a committed build (STP §13).
Status	Pass

TC-023 — Member role change and removal

Field	Value
Requirements	FR-049
Type / Priority	Functional, system, positive + negative / Medium
Preconditions	Board B with Owner O and Editor members E and M, all connected in browsers.
Test data	—
Steps	1. O changes E to Viewer; E tries to draw. 2. O changes E back to Editor; E draws. 3. O removes M. 4. M reloads the dashboard. 5. E (Editor) tries to change O's or M's role through the API.
Expected result	Step 1: E's tools are disabled without reload and E's attempted edit (sent with `submitRaw`) is rejected with `FORBIDDEN`. Step 2: E can draw again. Step 3: M receives `access:revoked` (member removed) and is disconnected. Step 4: B is not on M's dashboard. Step 5: 403 `FORBIDDEN`.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Replaces the role-change part of TC-018 v0.1 (TC-018 now covers ownership transfer only).
Actual result	access.test.js and E2E: Editor → Viewer disables tools without reload and the next edit is FORBIDDEN; back to Editor; removed member receives MEMBER_REMOVED and is disconnected, board gone from dashboard; Editor PATCH → 403. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-024 — Restore: failure before commit, interrupted restore recovery and idempotent retry

Field	Value
Requirements	FR-033
Type / Priority	Functional, integration (fault injection) + recovery / High
Preconditions	Test build with the `FAULT_INJECT` hook (SAD §4.1.6); board B with TB-STD content saved as version n, later changed and versioned again; Owner O as scripted client; observer client V; access to the test MongoDB.
Test data	TB-STD; idempotency keys K1–K8 (UUIDs); a second retained version m; fault points `commit_write_error`, `after_pending_restore`, `after_version_insert`, `during_object_replace`, `after_complete`, `db_error_after_commit`.
Steps	1. Set `commit_write_error` (the journaled write of `pendingOp` fails). O restores version n with key K1. Clear the fault; O retries with K1. 2. Set `after_pending_restore`; O restores n with key K2 (the server stops right after the commit). Restart the server. 3. Set `after_version_insert`; O restores n with key K3 (the server stops after inserting the restore version). Restart. 4. Set `during_object_replace`; O restores n with key K4 (the server stops after replacing part of the objects). Restart; restart a second time without any request. 5. O repeats the restore of n with key K4; then sends a restore of version m with key K4. 6. Set `after_complete`; O restores n with key K5 (the server stops after the completing update, before any response or broadcast). Restart; O retries with K5. 7. Set `db_error_after_commit` (database writes fail after the commit; the process keeps running); O restores n with key K6; V submits an operation. Clear the fault and restart; O retries with K6. 8. Without faults, O sends a restore of n with key K7 and a restore of m with key K8 concurrently while V submits operations. 9. After every restart and request, compare the board objects with the restored version, `boards.seq` with the newest version's `boardSeq`, check that `pendingOp` is absent, count the restore versions, record the version numbers and `seq` values returned, and read the log.
Expected result	Step 1: 500 `INTERNAL`; board, `seq` and version list unchanged; no `board:state` to V; `version.restore_failed` logged; the retry with K1 then succeeds with 201 and exactly one new "restore of n" version (the failed attempt recorded no key). Steps 2–4: O received no 201 and V no `board:state` before the crash; after the restart the board equals version n, exactly one new "restore of n" version exists for that key with `boardSeq = boards.seq`, `pendingOp` is absent and `version.restore_rolled_forward` is logged; the second restart in step 4 changes nothing. Step 5: the retry returns 201 with the same `newVersion`, `restoredFrom` and `seq` as the completed restore and no additional version; the request for version m returns 409 `IDEMPOTENCY_KEY_REUSED` and changes nothing. Step 6: after the restart nothing is rolled forward (no `version.restore_rolled_forward`), the board equals n, and the retry with K5 returns the recorded result without a new version. Step 7: O receives 500 `INTERNAL` (no success); `recovery.required` is logged; V's operation is rejected with `INTERNAL`; after the restart the restore is complete as in step 2 and the retry with K6 returns the recorded result. Step 8: both requests succeed with consecutive version numbers and different `seq` values; the board equals the version restored by the request answered last; every operation of V is applied entirely before or after each restore, as shown by the `seq` order.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Requires the `FAULT_INJECT` hook (test builds only; implemented, historical v1.0 note: planned when this was written).
Actual result	server/tests/integration/recovery.test.js with TB-STD as version n: steps 1–7 with every listed fault point (server killed and restarted as a child process), and step 8 in versions.test.js; board equals n, one restore version per key with boardSeq = boards.seq, pendingOp absent, retries return the recorded result, key reuse 409. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-025 — Clear: version write failure and interrupted clear recovery

Field	Value
Requirements	FR-007
Type / Priority	Functional, integration (fault injection) + recovery / High
Preconditions	Test build with the `FAULT_INJECT` hook; board B with at least 50 objects and changed content since its latest version; Owner O and observer V as scripted clients; access to the test MongoDB.
Test data	Fault points `version_write_error`, `after_preclear_version`, `after_pending_clear`, `during_object_delete`, `after_complete`, `db_error_after_commit`, `recovery_db_error`.
Steps	1. Set `version_write_error` (the before-clear version write fails); O sends `board:clear`. 2. Set `after_preclear_version` (the server stops after writing the before-clear version, before the commit); O sends `board:clear`. Restart; O and V reconnect. 3. Set `after_pending_clear`; O sends `board:clear` (the server stops right after the commit). Restart; O and V reconnect. 4. Add 50 objects; set `during_object_delete`; O sends `board:clear` (the server stops after deleting part of the objects). Restart; restart a second time without any request. 5. O sends `board:clear` again on the now empty board. 6. Add objects; set `after_complete`; O sends `board:clear` (the server stops after the completing update, before the acknowledgement and broadcast). Restart. 7. Add objects; set `db_error_after_commit`; O sends `board:clear`; V submits an operation; then restart the server. 8. Add objects; set `after_pending_clear`; O sends `board:clear`. Restart with `recovery_db_error` (database writes fail during start-up recovery); then restart without the fault. 9. After each step check the board objects, `boards.seq`, `pendingOp`, the newest versions, the acknowledgements and broadcasts received by O and V, and the log.
Expected result	Step 1: rejected with `INTERNAL`; board content and `seq` unchanged; V receives no `op:applied`; `op.clear_failed` logged. Step 2: after the restart the board still holds its content, `seq` is unchanged, `pendingOp` is absent and nothing is rolled forward; the extra "before clear" version (if any) equals the current content. Steps 3–4: O received no acknowledgement and V no `op:applied` before the crash; after the restart the board is empty, `boards.seq` is one higher than before the clear, `pendingOp` is absent, a "before clear" version holds the pre-clear content, `board.clear_rolled_forward` is logged, and the reconnected clients show the empty board; the second restart changes nothing. Step 5: acknowledgement `noop: true`; no new version; `seq` unchanged. Step 6: after the restart nothing is rolled forward; the board is empty with the `seq` of the completed clear; O never receives a success acknowledgement and the client shows the "unconfirmed changes" notice. Step 7: no success acknowledgement for the clear; `recovery.required` logged; V's operation rejected with `INTERNAL`; after the restart the clear is complete as in step 3. Step 8: the first restart logs `recovery.required` and the server exits with a non-zero status without accepting connections; after the second restart the clear is complete as in step 3.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Requires the `FAULT_INJECT` hook (test builds only; implemented, historical v1.0 note: planned when this was written).
Actual result	server/tests/integration/recovery.test.js steps 1–8 with 50 objects (steps 6–8: 5 objects): version_write_error rejected; crashes before/after the commit rolled back/forward as specified; second restart changes nothing; noop clear; unconfirmed-changes notice; recovery_db_error exits non-zero and the next start completes the clear. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

A.2 Security test cases

A security requirement counts as verified only when its test cases have been executed with status Pass (RTM status P); documenting a test does not verify anything. Execution 12-10-2026: TC-101 to TC-108 Pass; TC-109 Blocked (no deployment), so SEC-REQ-007 remains unverified.

TC-101 — Crafted real-time messages from unauthorised roles

Field	Value
Requirements	FR-051, SEC-REQ-003
Type / Priority	Security, integration (scripted Socket.IO clients), negative / High
Preconditions	Board B (Owner O) with Viewer member V and a Viewer guest G; board B2 owned by another user, on which V has no role; observer client C (Editor) joined to B.
Test data	From V and G: `object:add`, `object:update`, `object:delete`, `board:clear`; the same with an extra payload field `role: "owner"`; `op:submit` carrying B2's `boardId`; `board:join` for B2. At least 12 attempts.
Steps	1. Send each crafted message from V's and G's connections. 2. Record every acknowledgement. 3. Inspect C's received events and the database contents of B and B2 after all attempts. 4. Inspect the security log.
Expected result	Every attempt is rejected (`FORBIDDEN` on B; `BOARD_NOT_FOUND` for B2); C receives no `op:applied` for any attempt; B and B2 unchanged; each attempt logged as `authz.denied`.
Pass/fail criteria	Pass only if 100% of attempts are rejected and nothing is stored or broadcast.
Dependencies / limitations	—
Actual result	server/tests/integration/access.test.js: 16 crafted messages from Viewer and guest (incl. role:'owner' payloads, foreign boardId, foreign board:join) all rejected; observer received no op:applied; boards unchanged; each logged authz.denied. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-102 — REST authorisation matrix and object references

Field	Value
Requirements	FR-033, FR-044, FR-051, SEC-REQ-003
Type / Priority	Security, integration (API), negative + control / High
Preconditions	Board B with Owner O, Editor E, Viewer V, guest G; registered user X with no role on B; board B3 where G has no access.
Test data	Owner-only endpoints: delete board, create/list/revoke links, change/remove member, transfer ownership, restore version. Viewer endpoints: get board, list members, list/preview versions. Every restore call carries a valid, new `Idempotency-Key`, so that a rejection reflects authorisation and not a missing header.
Steps	1. Call every Owner-only endpoint as E, V and G. 2. Call every Owner-only and Viewer endpoint as X. 3. Call every Viewer endpoint for B3 with G's guest token. 4. Call each endpoint as the permitted role (control). 5. Inspect the database and version list after steps 1–3.
Expected result	Step 1: all `403 FORBIDDEN`. Steps 2–3: all `404 BOARD_NOT_FOUND` with no board data in the response (no-role rule, SAD §3.9.3). Step 4: all succeed. Step 5: no data changed and no version created by rejected calls.
Pass/fail criteria	Pass only if 100% of unauthorised calls are rejected as specified and every control call succeeds.
Dependencies / limitations	—
Actual result	access.test.js: Owner-only endpoints as Editor/Viewer/guest → 403; every endpoint as non-member → 404 with no board data; guest token on another board → 404; data and versions unchanged; controls succeed; 40 authz.denied entries. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-103 — Share-link revocation and expiry

Field	Value
Requirements	FR-047, FR-048, SEC-REQ-004
Type / Priority	Security + functional, system / High
Preconditions	Owner O on board B; Editor link L1 (7 days) and Viewer link L2 with expiry shortened to 60 s by `LINK_EXPIRY_OVERRIDE_SEC`; guest G1 joined via L1, guest G2 via L2; registered user R became a member via L1.
Test data	—
Steps	1. O revokes L1; measure the time until G1 is disconnected. 2. Open L1 in a new browser. 3. Check R's access to B. 4. Wait until L2 expires; measure the time until G2 is disconnected. 5. Open L2 in a new browser. 6. G1's client retries the connection with its old guest token.
Expected result	G1 receives `access:revoked` and is disconnected within 5 s; L1 → "link revoked" (`410 LINK_REVOKED`); R still has access; G2 is disconnected within 5 s after expiry; L2 → "link expired" (`410 LINK_EXPIRED`); step 6: the reconnection is refused and `board:join` is rejected (no stale guest permission).
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Uses a test-build override for the 60 s expiry; production expiry options are 1 h, 24 h and 7 days.
Actual result	access.test.js: revoked link → guest disconnected < 5 s, 410 LINK_REVOKED, member keeps access, old guest token refused at the handshake; expiry with LINK_EXPIRY_OVERRIDE_SEC=3 → guest disconnected < 5 s with LINK_EXPIRED notice, 410 LINK_EXPIRED. E2E revocation notice in the browser. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-104 — Link token strength, storage and secret-free logs

Field	Value
Requirements	SEC-REQ-004, SEC-REQ-008
Type / Priority	Security, integration + inspection / High
Preconditions	No share links in the test database; log file `logs/server.log` cleared; the log schema of SAD §4.4 available to the scan script.
Test data	1,000 generated links.
Steps	1. Generate 1,000 links; check token length, alphabet and uniqueness. 2. Inspect `shareLinks` in the database. 3. Run TC-101 to TC-103 and TC-105 to TC-108; in addition, attempt one login with an unregistered e-mail and one with a registered e-mail and a wrong password, and as Owner change a member's role, transfer ownership, restore a version and delete a test board. 4. Parse every log line as JSON and check each security entry against the field rules of SAD §4.4. 5. Search the log for every generated password, session token, guest token, link token, the `Authorization` header values, the entered e-mail addresses and a known board text string.
Expected result	Tokens are ≥ 22 base64url characters (design: 43) and unique; the database holds only `tokenHash` values. The log contains at least one entry for each of `auth.login_failed`, `authz.denied`, `link.created`, `link.revoked`, `member.role_changed`, `board.ownership_transferred`, `version.restored` and `board.deleted`; every security entry has `event`, a UTC `timestamp`, a non-empty `requestId`, `outcome`, and `reason` (an error code for rejections, `null` for successes); both failed logins have `actorType: "anonymous"`, `actorId: null`, `boardId: null` and `reason: "UNAUTHENTICATED"`, with `subjectId` set to U's internal account ID only for the registered e-mail and `null` for the unregistered one; role-change and ownership events carry the affected member's account ID as `subjectId`; board events carry the board's ID and the acting user's or guest's ID. The search finds no password, token, `Authorization` value, entered e-mail address or board text.
Pass/fail criteria	Pass if all checks succeed.
Dependencies / limitations	Runs after the other security tests so that the log contains their events.
Actual result	access.test.js: 1,000 tokens of 43 base64url characters, unique, only SHA-256 hashes stored. server/tests/integration/securityLog.test.js (LOG_LEVEL=debug, separate run instead of after the other security tests): every listed event type present with the required fields; failed logins anonymous with subjectId only for the registered e-mail; no password, token, Authorization value, entered e-mail or board text in any log line. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-105 — Password policy and storage

Field	Value
Requirements	SEC-REQ-001
Type / Priority	Security, integration + inspection, boundary / High
Preconditions	Test database available.
Test data	Passwords of 7, 8, 72 and 73 characters, generated at run time.
Steps	1. Register with each password length. 2. Register two accounts with the same password; inspect both user records. 3. Search the log for the passwords.
Expected result	7 and 73 rejected; 8 and 72 accepted; records contain bcrypt hashes that differ for identical passwords (salts) and no plain text; passwords absent from logs.
Pass/fail criteria	Pass if all checks succeed.
Dependencies / limitations	—
Actual result	server/tests/integration/auth.test.js: 7 and 73 characters rejected, 8 and 72 accepted; bcrypt hashes differ for identical passwords; no plain text in records or logs. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-106 — Authentication tokens: missing, tampered, expired, logged out, wrong board

Field	Value
Requirements	SEC-REQ-002
Type / Priority	Security, integration (API + Socket.IO), negative / High
Preconditions	Registered user U with board B; guest token GT for board B; board B4 owned by U; test JWT secret available to sign expired tokens.
Test data	No token; session token with one changed signature character; expired session token; session token after logout; expired guest token; GT used for B4.
Steps	1. Call `GET /boards` and `GET /boards/{B}` and connect to `/board` with each invalid session-token variant. 2. Log in, log out, then reuse the old token on both interfaces. 3. Use the expired guest token for `GET /boards/{B}` and `board:join`. 4. Use GT for `GET /boards/{B4}` and `board:join` on B4. 5. On an open socket whose token expires during the test, submit an operation after expiry. 6. Control: valid tokens.
Expected result	Steps 1–3: `401 UNAUTHENTICATED` / `connect_error UNAUTHENTICATED`; step 4: rejected (guest token not valid for B4) with no board data; step 5: operation rejected with `UNAUTHENTICATED`; step 6 succeeds.
Pass/fail criteria	Pass if every invalid variant is rejected on both interfaces and the controls succeed.
Dependencies / limitations	Uses the test JWT secret (`.env.test`) to create expired tokens.
Actual result	auth.test.js: missing, tampered, forged, expired, wrong-secret, alg none, logged-out tokens → 401 on REST and connect_error UNAUTHENTICATED; expired guest token rejected; guest token on another board → no data; open socket after logout/expiry cannot submit; controls succeed. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-107 — Input validation, XSS and oversized messages

Field	Value
Requirements	SEC-REQ-005
Type / Priority	Security, integration + browser, negative + boundary / High
Preconditions	Editor E connected to board B with a scripted client; observer O in a browser.
Test data	Missing `kind`; unknown `kind`; `color: "red"`; `strokeWidth` 0 and 21; text of 501 characters; non-numeric coordinates; unknown extra field; a message of 65,537 bytes (M1) and a valid message of at most 65,536 bytes as control; text object `<script>alert(1)</script><img src=x onerror=alert(2)>`.
Steps	1. Send each invalid payload as E. 2. Send the 65,537-byte message, then the control message. 3. Add the markup text object through the normal UI. 4. Observe O's browser (dialogs, console).
Expected result	Step 1 and the 65,537-byte message rejected (`VALIDATION_FAILED`, or connection-level rejection by the message limit) with the board unchanged; the control message accepted; in step 3 the text is displayed literally and no dialog or script runs.
Pass/fail criteria	Pass if every invalid input is rejected and no script executes.
Dependencies / limitations	Tests the message limit (M1) only; the board content limit (M3) is tested in TC-022.
Actual result	sync.test.js: every invalid payload VALIDATION_FAILED with the board unchanged; 65,537-byte frame closes the connection, 65,536-byte frame delivered; e2e/whiteboard.spec.cjs (Playwright, Chromium 1280 × 720; also passed in Chrome and Edge): markup text drawn literally, no dialog, no img element. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-108 — Rate limiting

Field	Value
Requirements	SEC-REQ-006
Type / Priority	Security, integration, negative / Medium
Preconditions	Registered user U; board B with scripted Editor clients F (flooder) and N (normal).
Test data	—
Steps	1. Attempt 6 failed logins for U within 15 minutes, then the correct password. 2. Attempt 6 failed logins for an unregistered e-mail address within 15 minutes; compare status codes and bodies with step 1, ignoring the per-request `requestId`. 3. F sends 100 operations within one second while N sends 2 per second. 4. Record F's and N's acknowledgements.
Expected result	The 6th failed attempt and the immediately following correct attempt return `RATE_LIMITED`; in step 2 attempts 1–5 return `401 UNAUTHENTICATED` with the same error code and message as for U and the 6th returns `429 RATE_LIMITED` with the same code and message as U's 6th attempt (only `requestId` differs), so the responses do not reveal whether the account exists; F's operations beyond 30 in that second are rejected with `RATE_LIMITED`; every operation of N is accepted.
Pass/fail criteria	Pass if all expected results hold.
Dependencies / limitations	Limits from D8b. Compares status, error code and message only; response timing of registered and unregistered addresses is not measured, so this test does not demonstrate timing equality.
Actual result	auth.test.js: 6th failed login and the following correct one RATE_LIMITED; identical status/code/message for the unregistered address; 100 operations in one second → exactly 30 accepted, 70 RATE_LIMITED, every operation of the other client accepted. Run 12-10-2026 on the uncommitted working tree of branch docs/setup (base commit 82af885); Windows 10, Node.js 26.7.0, MongoDB 8.2.1.
Status	Pass

TC-109 — Transport security in deployment

Field	Value
Requirements	SEC-REQ-007
Type / Priority	Security, manual inspection / High
Preconditions	A deployed (non-local) instance exists.
Test data	—
Steps	1. Open `http://<host>/` and observe the redirect. 2. Inspect the TLS connection details in the browser (protocol version). 3. Confirm the Socket.IO connection uses `wss://`. 4. Attempt a TLS 1.1 connection with a TLS client tool.
Expected result	HTTP redirects to HTTPS; protocol TLS 1.2 or 1.3; WebSocket uses `wss://`; the TLS 1.1 connection is refused.
Pass/fail criteria	Pass if all checks succeed on the deployed instance.
Dependencies / limitations	**No deployed instance exists.** Until one exists this test stays "Blocked" (reason: no deployment, §13) and SEC-REQ-007 remains unverified; the local test environment uses plain HTTP/WS by design.
Actual result	No deployed HTTPS instance exists (DECISIONS.md E3). FORCE_HTTPS/TRUST_PROXY are implemented but unverified. No step of this test was executed; status recorded 12-10-2026. 14-10-2026: the application-side controls were corrected — `FORCE_HTTPS` now requires `TRUST_PROXY` and an `https://` `PUBLIC_URL` (otherwise the server refuses to start instead of looping redirects), the redirect target comes from `PUBLIC_URL` instead of the request's `Host` header, real-time connections not forwarded as HTTPS are refused, and the client CSP allows only `wss:` — and were tested locally with a simulated TLS-terminating proxy (`server/tests/integration/transport.test.js`, 5 passed; `server/tests/unit/config.test.js`; uncommitted working tree, base commit 82af885; `docs/evidence/2026-10-14-full-verification/`). These local checks are not steps of this test: no TLS endpoint, certificate, TLS version or `wss://` connection to a deployment was tested, so the status stays Blocked. Later on 14-10-2026: steps 1–4 are automated by `scripts/verify-tls.js` (no account needed). Its local self-test (`server/tests/integration/tls.test.js`: the server behind a local TLS 1.2+ proxy with a throw-away self-signed certificate) passed all seven checks — 308 redirect to the configured HTTPS origin, HSTS, TLS 1.2 with the trusted certificate, TLS 1.1 refused by the server, Socket.IO over `wss://` reaching the application, plain `ws://` refused (`docs/evidence/2026-10-14-tls-local/`). The TLS settings checked there are the test proxy's, not a deployment's; status unchanged (Blocked). 15-10-2026: the local checks were repeated on the same code — self-test 7/7 PASS (including TLS 1.1 refused), and the transport and configuration tests passed, now with one more regression test: REST requests and real-time connections read a multi-valued `X-Forwarded-Proto` identically (first value, as Express does), shown to fail when the Socket.IO check is deliberately changed to the last value. Still no deployment; status unchanged (Blocked).
Status	Blocked